

Technetium

Tc (Tc I). Strongest classified transitions below 2 μm (NIST): 37 levels, 70 lines.

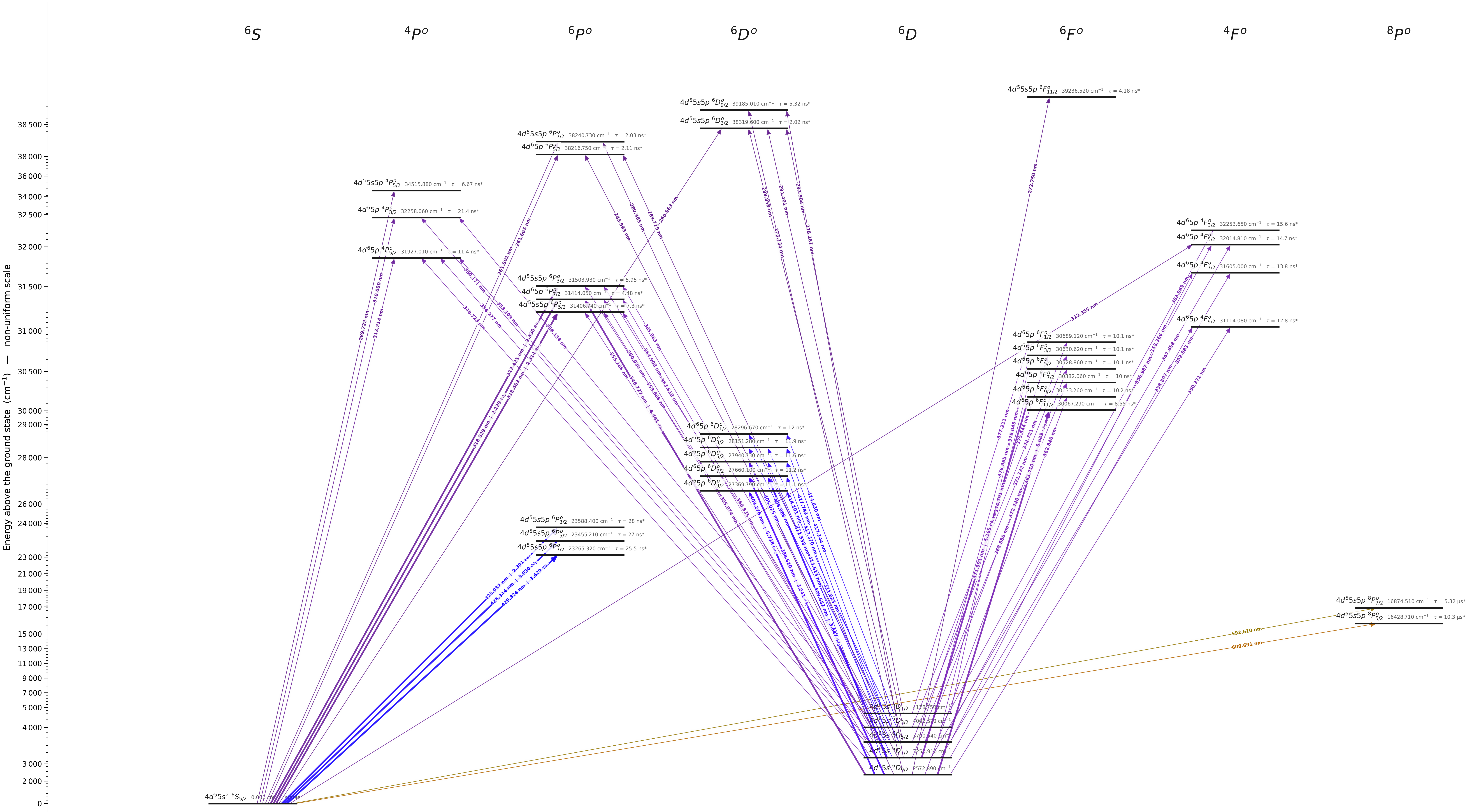
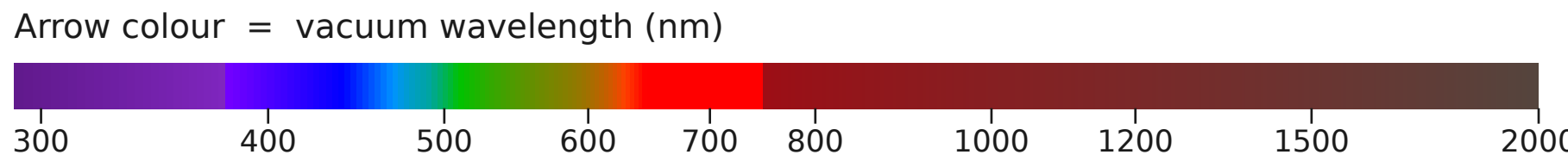
Tc⁺ ionisation limit 57421.68 cm⁻¹ = 7.11938 eV (174.150 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J\rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{3/2}$	423.9372	423.8178	707.1624	2.391	3.800e+07	5.682	100	first step of the three-step Ti:sapphire resonance-ioniza...
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{5/2}^o$	426.3445	426.2245	703.1695	3.030	4.000e+07	5.889	100	first step of the three-step Ti:sapphire resonance-ioniza...
$4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{7/2}$	429.8243	429.7034	697.4768	3.629	4.200e+07	6.239	100	first step of the three-step Ti:sapphire resonance-ioniza...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4d^45s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{3/2}^o$	423.937	97-99	308(14)
$4d^45s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{5/2}^o$	423.937	98-99	324(14)
$4d^45s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{5/2}^o$	426.344	97-99	313(14)
$4d^45s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{7/2}^o$	426.344	98-99	327(14)
$4d^45s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{7/2}^o$	429.824	97-99	347(14)
$4d^45s^2\ ^6S_{5/2} - 4d^55s5p\ ^6P_{7/2}^o$	429.824	98-99	369(14)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$

⁹⁹Tc hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)						
$4d^55s^2\ ^6S_{5/2}$	-107.34(8)	0.3(20)	F=2: +1476.06	F=3: +1153.95	F=4: +724.51	F=5: +187.76	F=6: -456.26	F=7: -1207.50	
$4d^55s5p\ ^6P_{9/2}$	799.99(26)	-44(4)	F=1: -15419.93	F=2: -13813.68	F=3: -11405.61	F=4: -8197.28	F=5: -4190.80	F=6: +611.23	F=7: +6205.67
$4d^55s5p\ ^6P_{3/2}$	894.08(28)	57(4)	F=2: -12267.29	F=3: -9602.27	F=4: -6042.22	F=5: -1181.38	F=6: +3787.40	F=7: +10072.75	
$4d^55s5p\ ^6P_{3/2}$	1183.27(27)	-13(3)	F=3: -9767.75	F=4: -5026.27	F=5: +892.70	F=6: +7983.92			

Measured A, B constants; sources in the data file.